

Tight Squeeze

Dean Ing

An abstract graphic design featuring a solid green background. Overlaid on this are several bright blue geometric shapes: a large inverted triangle in the bottom left, a solid circle in the bottom center, and a thick blue arc in the bottom right. In the upper half, there are two vertical blue lines and a large blue arc connecting them. In the lower half, there are two more vertical blue lines and a thick blue arc at the bottom right.

Project Gutenberg

Tight Squeeze

Dean Ing

An abstract geometric pattern on a green background. It features several blue shapes: a large inverted triangle in the bottom left, a large circle in the bottom center, and various curved lines and vertical bars. The text 'Project Gutenberg' is written in white over the bottom right portion of the pattern.

Project Gutenberg

The Project Gutenberg eBook of Tight Squeeze

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Tight Squeeze

Author: Dean Ing

Release date: November 5, 2007 [eBook #23337]

Language: English

Other information and formats: www.gutenberg.org/ebooks/23337

Credits: Produced by Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK TIGHT
SQUEEZE ***

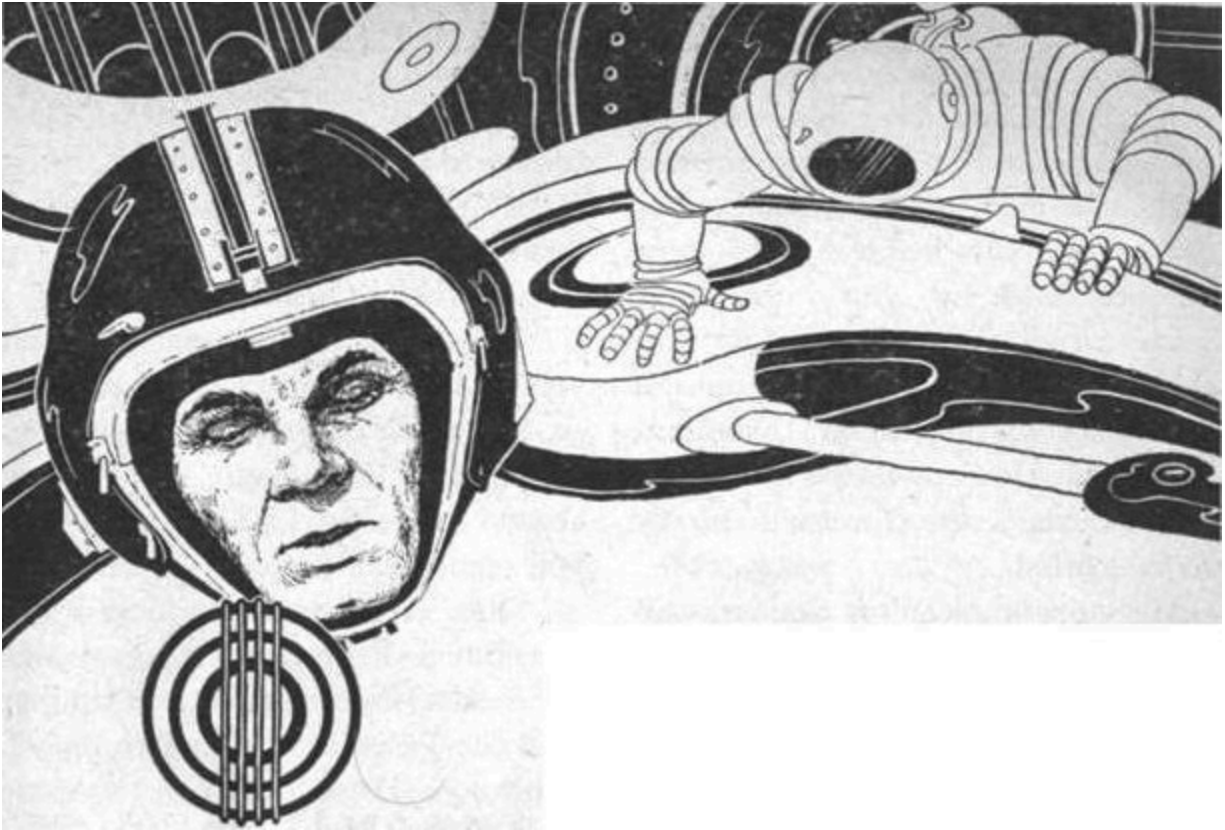
TIGHT SQUEEZE

BY DEAN C. ING

Illustrated by Freas

[Transcriber note: This etext was produced from Astounding Science Fiction February 1955. Extensive research did not uncover any evidence that the U.S. copyright on this publication was renewed.]

He knew the theory of repairing the gizmo all right. He had that nicely taped. But there was the little matter of threading a wire through a too-small hole while under zero-g, and working in a spacesuit!



MacNamara ambled across the loading ramp, savoring the dry, dusty air that smelled unmistakable of spaceship. He half-consciously separated the odors; the sweet, volatile scent of fuel, the sharp aroma of lingering exhaust gases from early morning test-firing, the delicate odor of silicon plastic which was being stowed as payload. He shielded his eyes against the sun, watching as men struggled with the last plastic girders to be strapped down, high above the dazzling ground of White Sands. The slender cargo doors stood open around *Valier's* girth, awaiting his own personal O.K.

This flight would be the fourth for Major Edward MacNamara; as he neared the great, squatting shock absorbers he could feel the tension begin to knot his stomach. He had, of course, been overwhelmed by the opportunity to participate in Operation Doughnut. The fact that he had been one of the best mechanical engineers in the Air Force never occurred to him at the time. He was a pilot, and a good one, but he had languished as C.O. of a maintenance squadron for nearly two years before he was given another crack at glory.

Now, he wasn't at all sure he was happy with the transition. They needed master mechanics for Operation Doughnut, but he felt they should be left on the ground when the towering supply rockets lifted.

He stopped, leaning against scaffolding as he saw a familiar figure turn toward him. He cupped his hands before his face.

"Hey, douse that butt! Can't you ... oh, Mac!" The commanding voice trailed off in a chuckle. Better to clown his way through the inspection, MacNamara thought, than to let Ruiz notice his nervousness. The co-pilot, Ruiz, walked toward him, still smiling. "One of these days, boy, you gonna go too far. Thought you were a real, eighteen carat saboteur." He clapped MacNamara on the shoulder and gazed aloft. "Good day for it. No weather, no hangover, no nothing."

"Yeah. You know, Johnny, I've been thinking about a modification for our breathing oxy." He sniffed appreciatively.

"What's that?"

"Put a little dust in it, a few smells. That stuff we breathe is just too sanitary!"

"I know what you mean. I sure begin to crave this filthy, germ-filled air after a few hours out there." They both smiled at the thought, then turned to the business at hand.

"By the way, Johnny, what're you doing out so early? Didn't expect to see you cabbies before ten."

"I donno," the bronzed Ruiz replied. "Went to bed early, woke up at six and couldn't drop off again. And here I am. Carl ought to be along around nine-thirty. Thought I'd help you preflight, if you want me to."

"Sure." He wanted nothing of the sort, but had the tact not to say so.

Edward MacNamara was as familiar with the *Valier* as he was with the tip of his nose. He had been on the scene when Dan Burke test-hopped the third stage, had made improvements and re-routing jobs, and had memorized every serial number of every bearing that went into *Valier*. As Flight Engineer, he was supposed to.

With Johnny Ruiz helping a little and hindering a little, he finished his tour of the cargo sections and grinned his approval to a muscular loading technician. "They can button her up, sergeant. I couldn't do a better job myself." It was a compliment of the highest order, and they both knew it.

Riding the tiny lift down to ground level, MacNamara stopped them every ten feet or so to circle the catwalks. He noticed Ruiz's impatience about halfway down. "No hurry, Johnny. I don't want another *Wyld* on our hands." He knew he shouldn't have said it, but it slipped out anyway. Everyone tried to forget the *Wyld* disaster, particularly the flight personnel. The *Wyld*, one of the first ships to be built, had made only two orbits before being destroyed. Observers stated that a cargo hatch had somehow swung open when the *Wyld* was only a thousand feet in the air. At any rate, the pilot reported damage to one second-stage fin and tried to brake his way down. The *Wyld* settled beautifully, tilted, then fell headlong. The resultant explosion caused such destruction that, had there not been a number of men in orbit and waiting for supplies, the project might have been halted, "temporarily." It was generally conceded that a more thorough preflight could have prevented the *Wyld's* immolation.

Ruiz was noticeably quieter during the remainder of the inspection. The external check completed, MacNamara strapped a small flashlight to his wrist and began the internal inspection, jokingly called the autopsy.

An hour and over a hundred and fifty feet later, MacNamara wheezed as he swung over the bulkhead at the base of *Valier's* third and top stage. His aching limbs persuaded him to take a breather. After all, his complete inspection of the day before really made a final preflight unnecessary, and passing near the frigid oxygen tanks was a day's work in itself. He listened to the innumerable noises around and below him. The clicks and hums near him meant that Ruiz, having given up following him, was checking out the flight controls, with power on only in the top stage. From below came a vibrational rushing noise, nearly subsonic, which told him of the fueling operation. He thought of the electrical relays governing the fuel input and

shuddered. He violently disliked the idea of having hot wires near fuel of any kind, and rocket fuel in particular.

MacNamara swept his light over his wrist watch. Fifteen after. Logan should be along soon, he thought, and hastened to finish checking the conduits, servos, pumps and hydraulic actuators below the cabin level. This done, he crawled up the final ladder to the cabin, or "dome."

"Well," cried a cheerful voice, "if it isn't our grimy Irishman."

MacNamara shook the sweat from his brow and muttered, "Irishman, is it? How about 'Logan'? That's a good Scandinavian name."

"How about Logan? He's great, as usual. Just look at me, Mac. What a specimen!" Logan, the inevitable optimist, bounced out of his acceleration couch and spread his arms wide as if to show the world what a superman he, Carl Logan, was. The gesture and its intimations made MacNamara smile. Logan wasn't much over five feet tall, and his flight suit made him look like a bald pussycat. His small physique covered a fantastic set of reflexes, however, and Logan's sense of humor was a quality of utmost importance. He hadn't an enemy in the world. His enemy was out of this world by definition; Logan wanted to conquer space and, so far, was doing just that.

"O.K., O.K. Laugh. Just remember this, Gargantua; I may not be tall, but I sure am skinny." MacNamara smiled again, nodding agreement. "Well, don't everybody talk at once. How is she, Mac?"

"With luck," answered MacNamara, "we might get ten feet off the turf." He paused for effect. "Seriously, Carl, she never looked better. You could take her up right now. Say, where's Johnny? I thought you'd just be checking in to the medics; looks like everybody's early today."

"He's probably over in some corner, making out his will. He was down below a while ago with a face a mile long."

Probably, thought Mac, *he's still thinking about the Wyld. Why did I have to bring that up?* Aloud, he said, "I ought to check the ground crew. Did you bring the forms?"

"Nope. Just my magnificent self. If anything had gone astray, they'd have told you."

"All the same, I think I'll go down and question the troops. Don't leave without me." He clambered out onto the catwalk, leaving the air lock open. The sun was riding higher every minute. In a little over an hour, he'd be a thousand miles away—vertically. The knot in his stomach began to form again. He wasn't scared, exactly; he kept telling himself "excited" was a nicer word.

The inspection forms signed, Mac held a short interrogation with the crew chief. The grizzled lieutenant, commissioned because of his long experience and responsibilities, gave *Valier* a clean bill of health. Each engine of the booster stage had been fired separately, before dawn. A cubic foot of mercury seemed to roll from Mac's shoulders as he saw Logan and Ruiz lounging at the bottom of the lift; there wasn't anything to worry about. He recalled feeling the tension before the other three flights, then chided himself. *Ya, ya, scared-y cat. Well, why not? It's a helluva risk every time you make a shot, in spite of all the propaganda. Hooey; if you didn't know everything's O.K., you wouldn't be getting ready to make the shot. Yeah, but you never can tell*——He stopped his inward battle and forced some spring into his step as he moved toward Logan and Ruiz.

"I've tried my best to abort this big bug, but I can't find anything amiss."

"That's Granny MacNamara for you," jibed Logan. "Always trying to find fault." He winked at Ruiz and rubbed his hands together. "Well—tennis, anyone?"

Mac knew without asking that Logan, for all his apparent indifference, had painstakingly gone over every phase of the flight, checking distribution, radar, final instructions from Operations, weather, *et al.* Ruiz, as usual, watched and took notes as Logan gathered data.

At minus fifteen minutes, the trio was in the dome, checking personal equipment, while outside, the scaffolding ponderously slid away, section by

section.

There was little time for soliloquies of *to go, or not to go*; within the quarter-hour, Captain Ruiz and Majors MacNamara and Logan would be in readiness for the final count-down. With the emergency bail-out equipment checked, the men busied themselves on another continuity test of the myriad circuits spread like a human neural system throughout the ship. All relays, servo systems and instrument leads were in perfect condition as expected, and the trio was settled comfortably in acceleration couches with minutes to spare.

Logan contacted Ground Control a few seconds after the minus-three minute signal, informing all and sundry that Gridley could fire when ready. MacNamara sighed, thinking that if Logan's humor wasn't exactly original, it was surely tenacious.

The ship was brought to dim half-life at minus one minute by Logan's agile fingers, and as the final countdown rasped in his headset, Mac felt his innards wrestle among themselves.

Valier bellowed her enthusiasm suddenly, lifting her eight thousand-odd tons from the ground almost instantly. Inside, her occupants grimaced helplessly as they watched various instruments guide tiny pointers across calibrated faces. Mac's throat mike threatened to crush his Adam's apple, weighing five times its usual few ounces. Of his senses, sound was the one that dominated him; an intolerable, continuous explosion from the motors racked his mind like tidal waves of formic acid. He forced himself to overcome the numbness which his brain cast up to defend itself. Then, as quickly as it had begun, *Valier* fell deafeningly silent; that meant Mach 1 was passed.

It was an eternity before stage one separated. The loss of the empty hulk was hardly felt as *Valier* streaked high over the Texas border. Ruiz, watching the radarscope, saw Lubbock slide into focus miles below. *Next stop, Fort Worth*, he thought. *I used to drive that in five hours*. The jagged line of the caprock told him they were well on their way to Fort Worth already.

The altimeter showed slightly over forty-two miles when stage two detached itself. Logan, in constant contact with White Sands, was informed that they were tracking perfectly as *Valier* arrowed over central Texas toward rendezvous at the doughnut. The exhausted lower stages were forgotten now; only the second stage was of any concern anyway. The radar boys tracked it all the way down, ready to detonate it high in the air if its huge 'chutes wafted it near any inhabited community.

The motors of stage three blasted for a carefully calculated few seconds, then cut out automatically. With the destitution of his weight, Mac felt his spirits soar also. They were almost in orbit, now, climbing at a slight angle with a velocity sufficient to carry them around Earth forever, a streamlined, tiny satellite.

After the first few moments of disorientation, rocket crews found that a weightless condition gave them, ambiguously, a buoyant feeling. Only the doughnut crew had really adapted to this condition, living as they did without the effects of gravity for hours at a time every day. The temporary "housing" was rotated for comfort of the crews during rest periods, but while moving the plates and girders of the giant doughnut into place, they had no such luxuries. For these men, weightlessness became an integral part of their activities, but the rocket crews were subjected to this phenomenon only during the few hours needed to rendezvous, unload the cargo, and coast back after another initial period of acceleration.

Hence, Mac felt a strange elation when he tapped his fingers on the arm of his couch and saw his arm float upward, due to reaction from the tap.

Against all regulations, Logan unstrapped himself and motioned his comrades to do the same. This unorthodox seventh-inning stretch was prohibited because it left the pilot's arm-rest controls without an operator, hence could prove disastrous if, through some malfunction, the ship should veer off course.

The autopilot functioned perfectly, however, and Logan trusted it to the point of insouciance. The three men lounged in midair, grinning foolishly as they "swam" about the tiny cabin. No more satisfying stretch was ever enjoyed.

A few minutes of this was enough. Ruiz was the first to gingerly pull himself into his couch and his companions followed. Not a word had passed between them, since they were at all times in contact with monitor stations spaced across the world below. The first time they had enjoyed this irregular horseplay, on the second trip, Logan had made the mistake of saying, "Race you to the air lock!", and was hard put to explain those words. Nor could Logan switch to "intercom only," since a sudden radio silence would create anxiety below. Only their heavy breathing would indicate unusual activity to Earthside.

They were nearing the intercept point, a thousand miles above the Atlantic, when they realized their predicament.

"I'm in a fix, Carl," said Ruiz, meaning that he had tentatively fixed a position of intercept. "Correct our elevation; we're point-nine degrees high."

"Right-o. Correction in five seconds from my mark—mark!"

For slight corrections in the flight path, small steering motors were utilized. These motors were located near the rear lip of *Valier's* conical cargo section on retractable booms. Extension of the motors with no resultant air friction gave a longer pivot arm and consequently better efficiency. Mac pressed the "Aux. Steer" stud and immediately three amber lights winked on in their respective instrument consoles.

Carl Logan fired the twelve o'clock motor briefly—only it didn't fire. The change in momentum wouldn't be much in any case, but it was always perceptible by feel and by instrument. There was no change.

Logan tried the firing circuit again, and again. Still *Valier* streaked along, now miles above the intended point of intercept. By this time, the embryo space station was quite near, sailing along in the 'scope beneath them. It slowly moved toward the top of the 'scope, passing *Valier* in its slightly higher relative velocity.

"We've got troubles, Mac—find 'em!" Logan had finally lost the devil-may-care attitude, but that fact was small consolation to MacNamara.

"Keep your mitts off those firing studs, Carl," he growled, unstrapping himself quickly. The malfunction was definitely in the auxiliary motor setup, he thought. A common trouble? It wouldn't pay to find out. If the other motors fired, it would only throw them farther off-course. If worst came to worst, they could roll *Valier* over and use the six o'clock auxiliary; there was a small arc through which the motors could turn on their mounts. But the trouble was unknown, and they might end up rifling or pinwheeling if they didn't let bad enough alone.

During his mental trouble-shooting, Mac was busily worming his bulk into a balloonish-looking suit identical to those worn by the doughnut's construction crew. Ruiz gave him some aid, helping him thrust his arms past the spring-folded elbow joints. For some reason, the legs gave less trouble. Within a fumbling few moments, he was ready for work.

He glanced at Logan through his visor, feeling a vicious pleasure over the beads of sweat on Logan's forehead. Time he sweated a little, thought the mechanic.

A final check of his headset followed, after which Mac oozed into the Lilliputian air lock at the bottom, now rear, wall of the cabin. He nodded to Ruiz, who secured the air lock, then adjusted his suit control to force a little pressure into his suit. Gradually the suit became livable. Then he cracked the other air-lock valve and allowed pressure to leak out around him.

His suit puffed out with soft popping noises and Mac heard the last vestige of air hiss out of the chamber. He found the hatchway too tight for comfort and had a moment of fear when his tool pack caught in the orifice, wedging him neatly. He could hear Logan and Ruiz through his earphones, explaining their plight to Ground Control. They wanted to know why in blue blazes *Valier* hadn't contacted the doughnut when it came within range, and Logan had no defense save preoccupation with his own plight. Belatedly, Ruiz made radio contact with the doughnut, which was still well within range. All this time, Mac busied himself with his inspection light, tracing the electrical leads to the small, turbine operated auxiliary motor fuel pumps.

"Mac?" Logan's voice startled him. "Can you brace yourself? I'm going to try to match velocities with the doughnut. Won't take over one 'g' for a few

seconds."

"Wait a minute." He looked wildly about him. *Valier* hadn't been built with a view toward stowaways; and every cubic inch of space was crammed with something, except for the passageway with its ladder, leading up from the main motor section. Well, if it wasn't over a "g," he could hang on to the ladder. Suit weighs another fifty pounds, though. My weight plus fifty, he thought. "Give me a chance to get set," he said aloud. He hooked one bulbous leg over a ladder rung and braced the other against a lower rung, hugging the ladder with both arms. "Any time you say, but kill it if you hear me holler!"

"Then five seconds from my mark—mark!" Mac tightened his grip, and then sagged backward as the main motors fired. The vibrations shook him slightly but deeply, and he fought to keep his hold. He felt his back creak and pop with the sudden surge of weight. Then the motors shut off, and Mac skidded several feet up the ladder. No matter how fast a man's reactions were, they couldn't be applied quickly enough to keep him from starting an involuntary leap after bracing against a suddenly removed gravity load. "All over, Mac. You O.K.?"

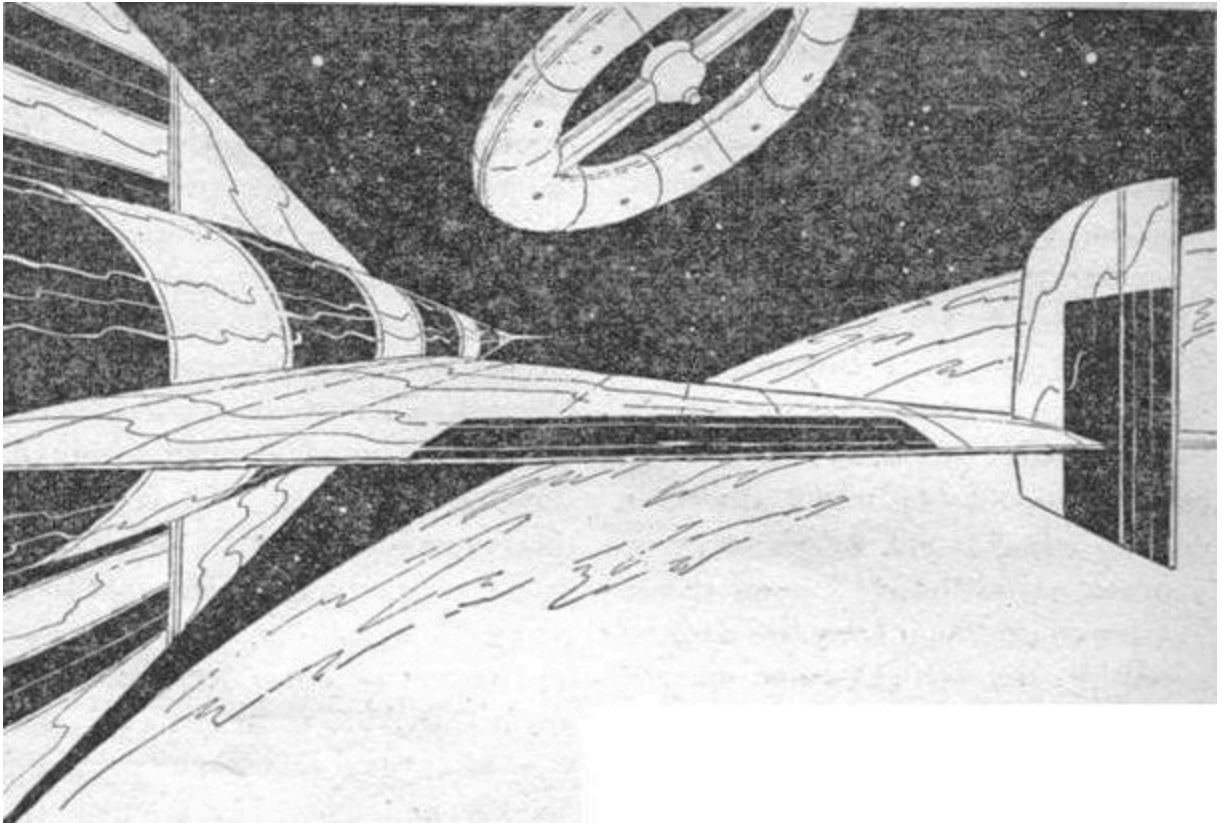
"Guess so, but I feel like a ping-pong ball. How're we sittin'?"

"Just fine," Ruiz cut in. "Find anything?"

"Not yet." Mac started his search anew. Everything seemed in perfect order up to the turbine pumps. Then, he feared, the trouble was near the little motors. That was tough, really tough. With the motors retracted it was next to impossible to get to them, past their hydraulically operated booms and actuators. Extended, he'd have to go outside. He cringed from the thought, although he knew that there was little to fear if he linked himself to the ship.

He peered along the beam of light, searching for some telltale discoloration in wiring, or a gleaming icy patch which would indicate a fuel leak. "Might be the firing plugs," he muttered.

"Let's hope not. Where are you, Mac? Maybe you better give us a blow-by-blow." Logan sounded worried.



"Good idea. Right now I'm at the nine o'clock actuator. Nothing so far." He looked around himself, forgetting for the moment how he was supposed to get past the equipment to the other auxiliary motor stations.

"Johnny," he said slowly, "I think you'd best break out the tapes. Auxiliary motor system; you'll find them under power plant." Months before, MacNamara had made a complete set of tape recordings of his own voice, recorded as he made a thorough-going rundown of every system and its components. This was a personal innovation which his fellow flight engineers considered folly. Extra weight, they scoffed. Undue complication. Mac nodded and went on with his impromptu speechmaking; a professional psychiatrist might have said, correctly, that Mac felt an unconscious need for supervision, a forgivable deficiency dating back to his cadet days. Mac simply claimed that the best of men could forget or omit when alone with a

few million dollars' worth of Uncle's equipment. This way he could remind himself of each step to be taken ahead of time, in his own way.

The co-pilot rushed to comply. Mac, waiting, suddenly remembered how to get past his obstacle. Internal braces which helped keep the tanks rigidly in place on Earth were of little use while in "freeloading," or gravity-less, state. The braces were removable, and Mac had loosened a single wing-nut to let the brace swing loose when he heard Johnny Ruiz's answer.

"Ready with your tape, Mac. Where shall I start it?"

"Run it through 'til you get to a blank spot, then another, then stop it." He was certain he didn't really need the tape, but it was a maintenance aid and he was determined to use it.

He heard a click, then a hum, as the recorder was jacked into his headset circuit. Immediately, a familiar voice began a slow dissertation on power leads from the dome, speeded up in the space of a second or two to a high-pitched alien gibberish, then to a faint scream. He began squirming around the turbine tanks, got past the first brace, and turned to attach it again. Of course it wasn't necessary, but—"PLAY IT SAFE" was embroidered on his brain by years of maintenance experience; back in his old maintenance squadron, he'd been called "the old lady" instead of "the old man," due to his insistence on precautions.

Ruiz slowed the tape suddenly, on cue, and Mac heard himself saying, "... Brace back in its slot and pin it. Be careful of those linkages on the turbine pumps. Now crawl around to the next brace and unpin it." Pause, scraping noises, and a muttered oath. "Pin sticks, but it won't without a load on it." It didn't.

He worked slower than he had on the ground, fumbling with the heavy gloves and cursing mightily. His voice rambled on, warning him of obstacles and reminding him about minor points that could give trouble. He listened carefully, discarding each suggestion.

Floating near the twelve o'clock auxiliary, Mac peered at each tubing connection, tugging and twisting. "Wait a minute," he said. His light flashed out at the motor, riding perched on its swivel, limned against cold, hard points of light that were the stars. His heart gave a bound. "I think I've

found it!" His other voice droned on morbidly. "Turn that thing off a minute, Johnny. Listen; there's a lead to the twelve o'clock fuel valve solenoid that looks like ... yes, I'm sure of it. It's pulled away from a bracket and looks like it might be charred." Mac twisted around to view the wiring better.

"Can you fix it?"

"Oh, sure, if that's all there is wrong. But I'd rather do the work with the motors retracted. Tell you what; retract them about forty-five degrees when I give the word."

Mac judged the distance the booms would cover during semiretraction and half floated, half crawled out of the way. He found himself breathing heavily, despite the freeload conditions. His suit was simply too cumbersome. The thought came to him that he didn't even know how long he'd been out of the dome. His breathing oxygen gauge showed half empty, so he must have been on the job for around a half hour. He rationed his supply a bit, hoping he could finish the job without a refill.

"O.K., Johnny, you can run the tape again. And retract the motors while you're at it." He heard the tape start again on its course, watching the booms.

They leaped inward, then, and Mac felt a crushing blow across his back. He shook his head groggily and yelled.

He tried to scramble from his place between motor and turbine fuel lines without success; he was trapped like a wild animal by the heavy actuator which had swung past his head. He heard himself say, "And be sure to stay clear of the actuator. It swings through a ninety-degree arc when it's operated."

"Oh, shut up! I know it; I just judged it wrong." The tape moved on unperturbedly, reminding him to inspect the actuator bearings and extension rods.

"Mac," came Logan's voice, "you might try to hurry it. If you can't get it fixed in an hour or two, we'll have to try rolling *Valier* down to the doughnut. But it's up to you, fella. Take your time."

"Well, you might help me a bit by raising this hydraulic unit offa my shoulders. Lucky it didn't squash me." The actuator stayed where it was. "Johnny! Carl! Do you read me?" No answer. Obviously, the actuator had smashed his transmitter, but left the receiver section intact. Then all he could hope for would be a suspicion from one of the others that all was not well. If they asked him any questions and he failed to reply, they'd figure something was wrong. Well, he couldn't count on that.

He struggled with his vulcanized suit, trying to squeeze from under the actuator. If I'd had them retract it completely, he thought, I'd be a dead man. It was a tight squeeze, but he inched his way out of the trap by using every ounce of strength at his command. If his suit tore, he'd know it in a hurry.

Gasping for breath, Mac drew himself into a crouch and regarded the offending wire. His flashlight still operated, and he could see the heavy insulation which had been scraped away. No charring; then it must have been the extension rods that had scissored through the insulation. The wire hung together by a thread, the strands of metal severed completely. He groped for his tool kit, trying to ignore the voice in his headset.

"Well, that takes care of the actuators. Now for these dinky motors. The swivel mounts have to work without any lubricant, so look for indications of wear and—"

Mac cursed under his breath. He sounded so cocksure, so all-knowing. He felt like beating himself. His earlier self, who had blithely toured *Valier* trailing the microphone wires without any real premonition of trouble. It always happens to the other guy—Not this time, chum, he reminded himself.

The gloves were systematically foiling his attempts to withdraw the coil of wire at his side. The tool kit was the ultimate in maintenance work, compact and complete with extension handles for the cutters and wrenches. Everything was there, but practically impossible to use. His fingers finally closed over the wire; he jerked it out and with it the splice tool. The little

pliers caromed from the brace above him and sailed out toward the motor, beyond the ship. He watched, horrified, as the tool slowly cartwheeled away into space.

"All right," he muttered, "scratch one splice tool. It was also my only pair of pliers, but I'll manage." He knew he could use the wire cutters in a pinch. "In a pinch," he repeated. "Oh, that's a hot one. That's about all that's happened this trip, so far. Pinch me, pinch the wiring—What a pinch!"

Holding the roll of wire tightly in one hand, he grasped the cutters and pulled them from the kit with utmost care. He unrolled a foot-long section of wire and clipped it off, laying his flashlight in the tool kit so that it would shine out in front of him. He managed to attach the tiny splice lugs by pinching them with the cutters, then moved cautiously to the wire which still drooped from the jumble of machinery. "Drooped" wasn't precisely the word; actually the wire had been bent into its position and stayed that way.

As the harried major reached for the brace on which the wire had been bracketed, his tool kit vomited flashlight, wrenches and screwdrivers, leaving him in total darkness. His cursing was regular, now, monotonous and uninspired. There was another pencil light in the kit, snapped tightly to the case, and Mac reached for the whole business. The spare light was a maintenance problem in itself. Question: How to retrieve a fountain pen sized object, when it's held by a small snap and the retriever is encumbered by three pairs of arctic mittens?

Mac saw his errant flashlight out of the corner of his eye, its beam fastened on a collapsed screw driver while both swam sluggishly toward the inspection ladder. He located the pencil light and jerked it loose, holding the short wire and cutters in his other hand.

This, Mac knew, was the crucial point. If he could splice the wire hanging in front of him, *Valier* would once more be in perfect shape. He would have welcomed an extra hand or two, as he straddled a brace and shoved the tiny flash between his headpiece and shoulder fabric. The wire should be stripped, he knew, but he hadn't the tools. They were scarcely ten feet from

him, but could have rested atop the Kremlin for all the good they did him. He got most of the strands of one end of wire shoved into a splice lug, and called it good enough. It was like trying to thread a needle whose eye was deeper than it was wide, while in a diving suit, using the business end of a paintbrush to start the thread.

He withdrew one hand and searched the kit for friction tape. It might be mentioned that an insulating tape which would be adhesive at minus two hundred degrees centigrade yet keep its properties at plus one thousand, was the near culmination of chemical science. Silicon plastic research provided the adhesive, an inert gum which changed almost none through a fantastic range of temperatures and pressures. The tape Mac used to insure his connection had an asbestos base, with adhesive gum insinuated into the tape. He wrapped the wire tightly, then bound it to the brace. He noticed his visor fogging up and felt a faint, giddy sensation. Anoxemia! He let the tape drift as he reached for his regulator dial. *What a fool he was*, he thought, *to starve his lungs*. He turned the dial to emergency maximum and gulped precious liters of oxygen-helium mixture. The gauge showed a store of the gas which might possibly be enough to last him, if nothing else went wrong; perhaps ten minutes.

The pencil flash, mercifully, still rested in a fold of his shoulder joint fabric. The insulation tape floated near his waist; he grabbed it and stowed it between his knee and the brace, then reached once again for the wiring.

This time the splice went on without a hitch. He pinched the splice lug and taped the whole works feverishly. It was done; he had won. The trip back should take only a couple of minutes. Replacing the wire cutters in his kit, he held the pencil flash before him and started retracing his route.

He passed the twelve o'clock brace, pinned it in place again and saw one of his tools floating to the right of his head. He gathered it in and swept his tiny flash around in search of other jetsam from his tool kit. He collected a wrench and the skittish flashlight, started toward the last brace between him and the ladder, and felt his legs go limp. He wasn't particularly alarmed about it; his arms and vision failed him too, but his brain hadn't enough incoming oxygen to care much, one way or the other. The few remaining

feet seemed to lengthen into a sewerlike passageway, then vanished as did all else as his perceptions died.

MacNamara was not the sort to wonder about heaven or hell when he first awoke. He saw a faintly rounded ceiling, a soft yellow tint accentuating its featurelessness. "How the devil—", he began. His voice failed him.

"Hi, Mac." Logan's beaming face loomed over him. "You rugged character, you. Cold as a pickle an hour ago, and already you're askin' silly questions." He held up his hand as Mac started to speak. "I hear you thinkin'. 'How the devil did I get here, and where is here?' In reverse order, this is the most comfortable berth in the doughnut's facilities, and you got here courtesy of one Johnny Ruiz. Myself, I wouldn't have taken the trouble."

Mac grinned back at his pilot and cleared his throat. "Well, where is he? I wanta shake his hand, or give him half my kingdom, or something."

"You know Johnny; the shy type. He'll be along after a while. You know, I think he kinda likes you; when you quit transmitting out there, Johnny was like a cat on a hot skillet. Finally decided to go back and have a look for himself, but I told him you probably had a hot game of solitaire going. Anyway, he went back and found you asleep on the job, and lost a good ten pounds getting your fat carcass through the air lock." That was a job that must have taxed both Ruiz and Logan, but Mac held his silence. "And that was about the size of it. *Valier's* parked outside with some of the boys, good as ever. Come on, we'll sop up some coffee."

Mac swung himself up to a sitting position and realized dizzily that he was mother-naked. His ribs felt pulverized. "You guys sure mauled me up," he said accusingly.

"Unavoidable, my dear grease-monkey. You needed a little artificial respiration; I never was too good at that."

"Well, whoever did the job rates a prize of some sort," Mac answered, "but my ribs tell me he had more enthusiasm than practice."

Logan smiled his old familiar smile, relieved to find his engineer in joking spirits. "The credit again goes to Johnny. But," he added, "try not to be too hard on him. Try giving artificial respiration to a big lump like yourself sometime, without any gravity."

Mac digested this tidbit as he pulled on a fresh pair of coveralls. "O.K.," he said, standing on the foamex "floor." "How did he do it?"

"Strapped you into your couch face down and locked his legs around it. I didn't dare apply any g's. Come on," he finished, "you've managed to upset every timetable in the project. Johnny's shaking like a leaf, or was when I left him. A bulb of coffee will do us both a world of good."

"I'm sold," Mac grunted, zipping up a flight boot. "But there's something I'd like to do, first chance I get."

"Which is?"

"Which is jettison every last strip of tape I have in *Valier*. I tell you, Logan," he went on as they entered the recreation bar, "you'll never know how degrading it is to hear useless, insipid information offered to you when you're in a tight spot, knowing full well the voice is your own!"

THE END

*** END OF THE PROJECT GUTENBERG EBOOK TIGHT SQUEEZE

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project GutenbergTM License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

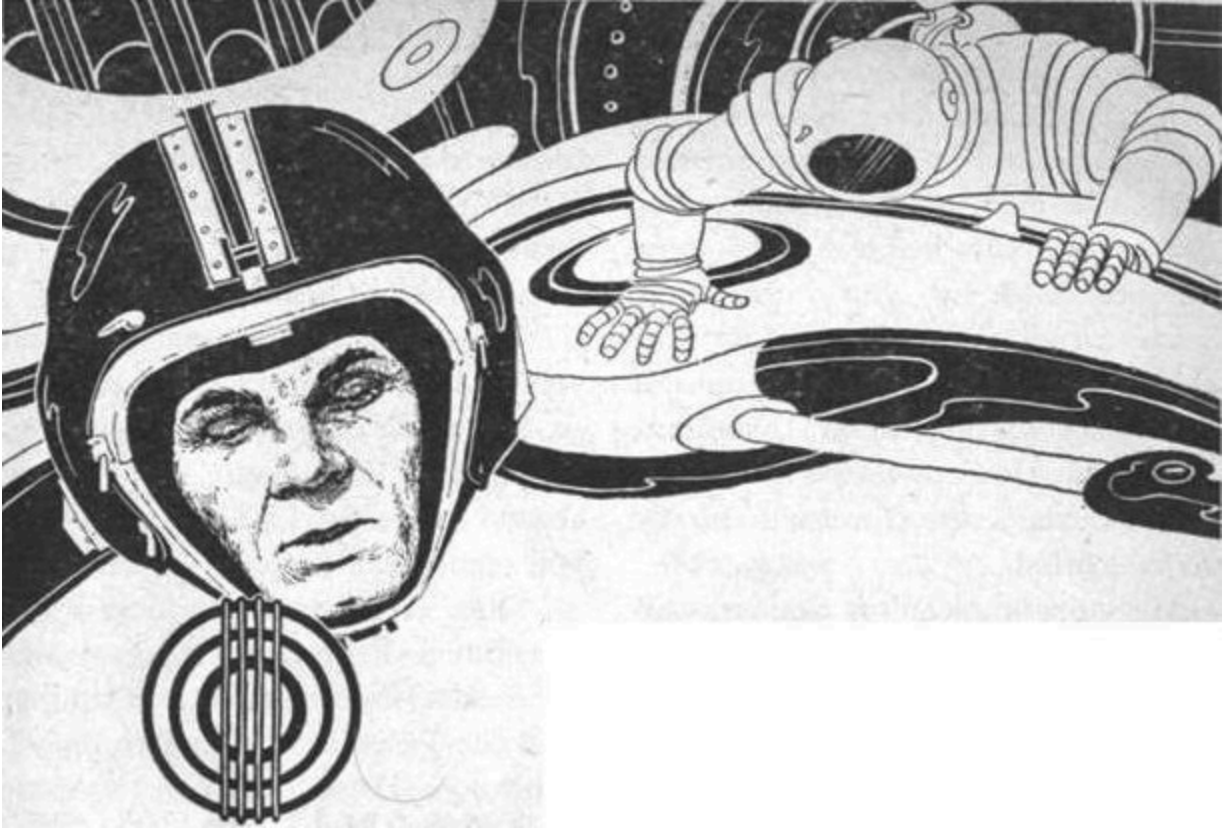
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

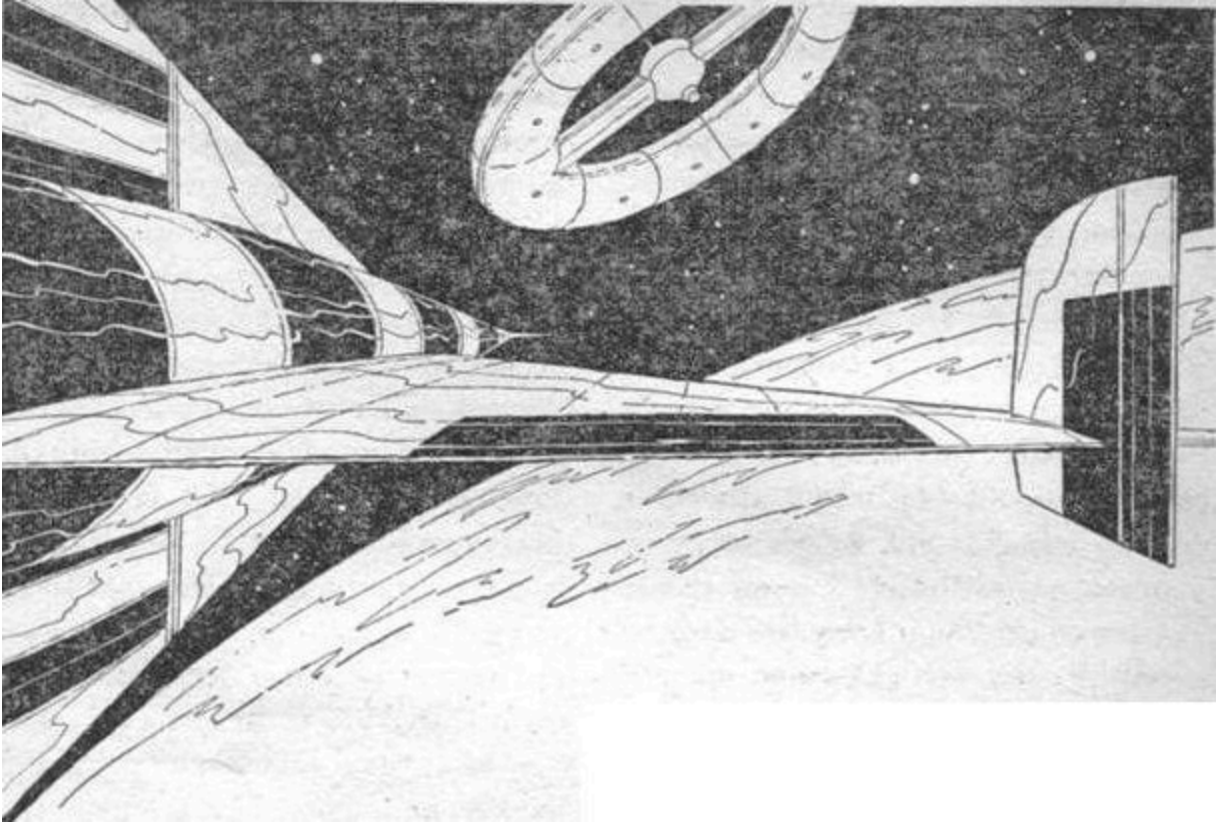
copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



[back](#)



[back](#)